

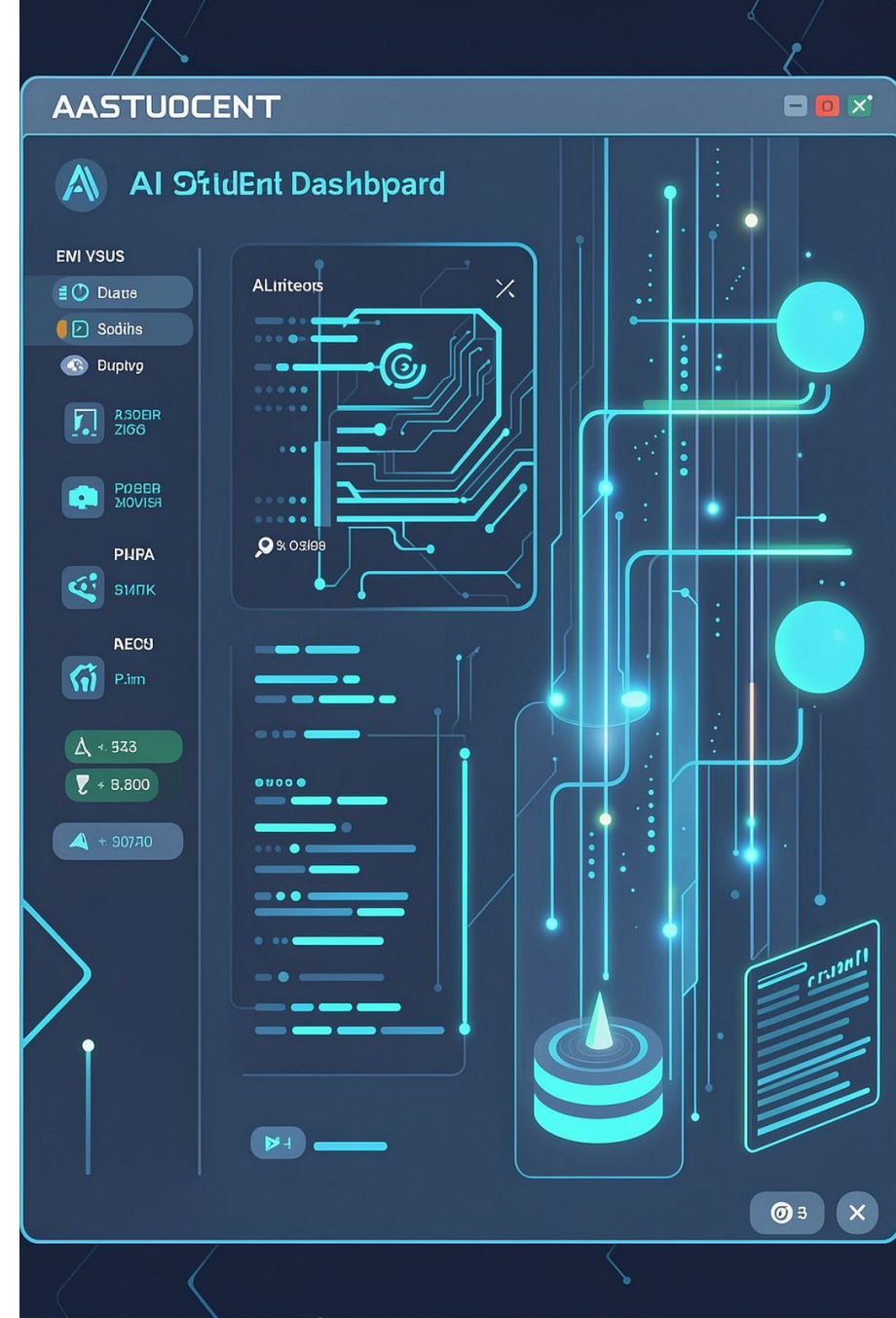
AI StudyBuddy

AI-Augmented Student Learning & Study Management Platform

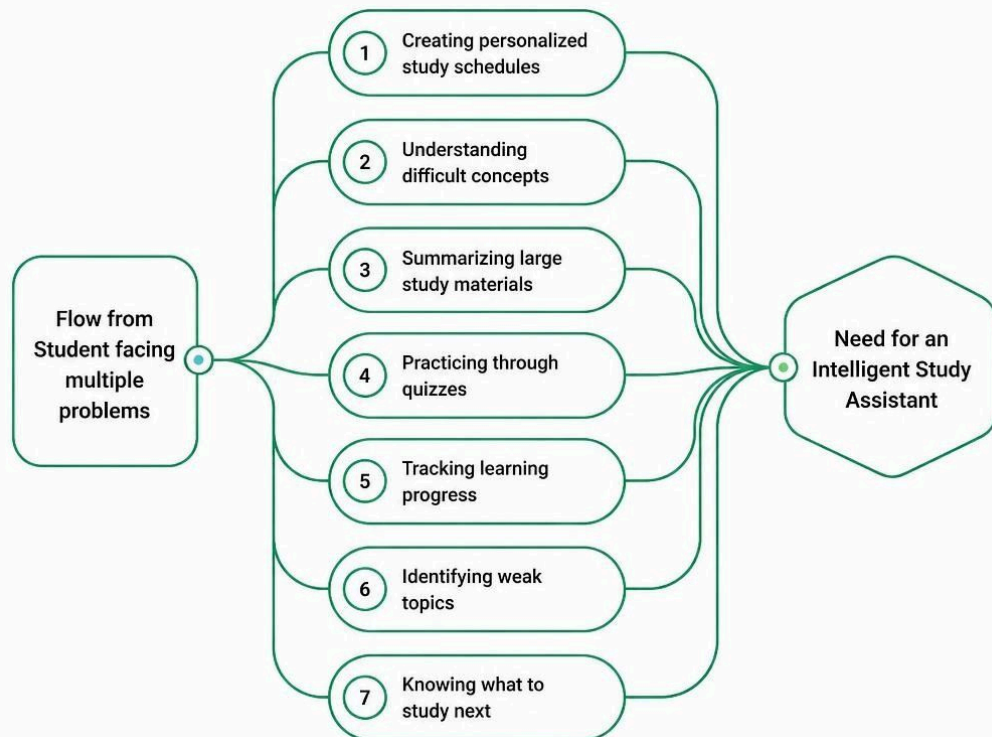
TN SKILL FINAL PROJECT

TEAM 11

Alan — Team Leader · Mathew · Karthikeyan · Kishore



The Challenge Students Face Every Day



The Core Gap

Traditional study tools provide information — but not a **personalized learning workflow**. Students are left without guidance on what to study, how to study, or how they are progressing.

- ❏ There is no unified platform that combines scheduling, summarization, quizzing, and AI-powered guidance in one place.

OUR SOLUTION

Introducing AI StudyBuddy

An intelligent, unified student learning platform powered by Google Gemini AI — combining planning, content understanding, practice, and progress tracking in one seamless experience.

Student Profile
Learning preferences & study habits

Progress & Analysis
Performance tracking & recommendations

AI Quiz Generator
Auto-generated practice quizzes



AI Study Planner
Personalized study schedules

AI Summarizer
Concise material summaries

AI Q&A
Instant concept explanations

What We Set Out to Build

1

Personalized Study Planning

Generate adaptive study schedules based on student profiles and goals.

2

Concept Understanding

Help students break down and understand difficult topics with AI explanations.

3

Smart Summarization

Reduce time spent on lengthy materials by converting them into concise summaries.

4

Automated Quizzing

Generate topic-based quizzes and score them automatically for instant feedback.

5

Progress Tracking

Continuously monitor learning progress and identify weak areas using AI analysis.

6

Unified Platform

Bring all study tools — planning, content, practice, and analytics — into one place.

What AI StudyBuddy Offers



Authentication

JWT-based secure login and registration



Student Profile

Learning preferences, education level, and study hours



Subjects & Topics

Organize and manage all learning content



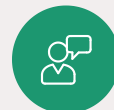
AI Study Planner

Generate personalized, adaptive study plans



AI Summarizer

Convert lengthy material into concise summaries



AI Q&A

Ask questions and receive clear explanations



AI Quiz Generator

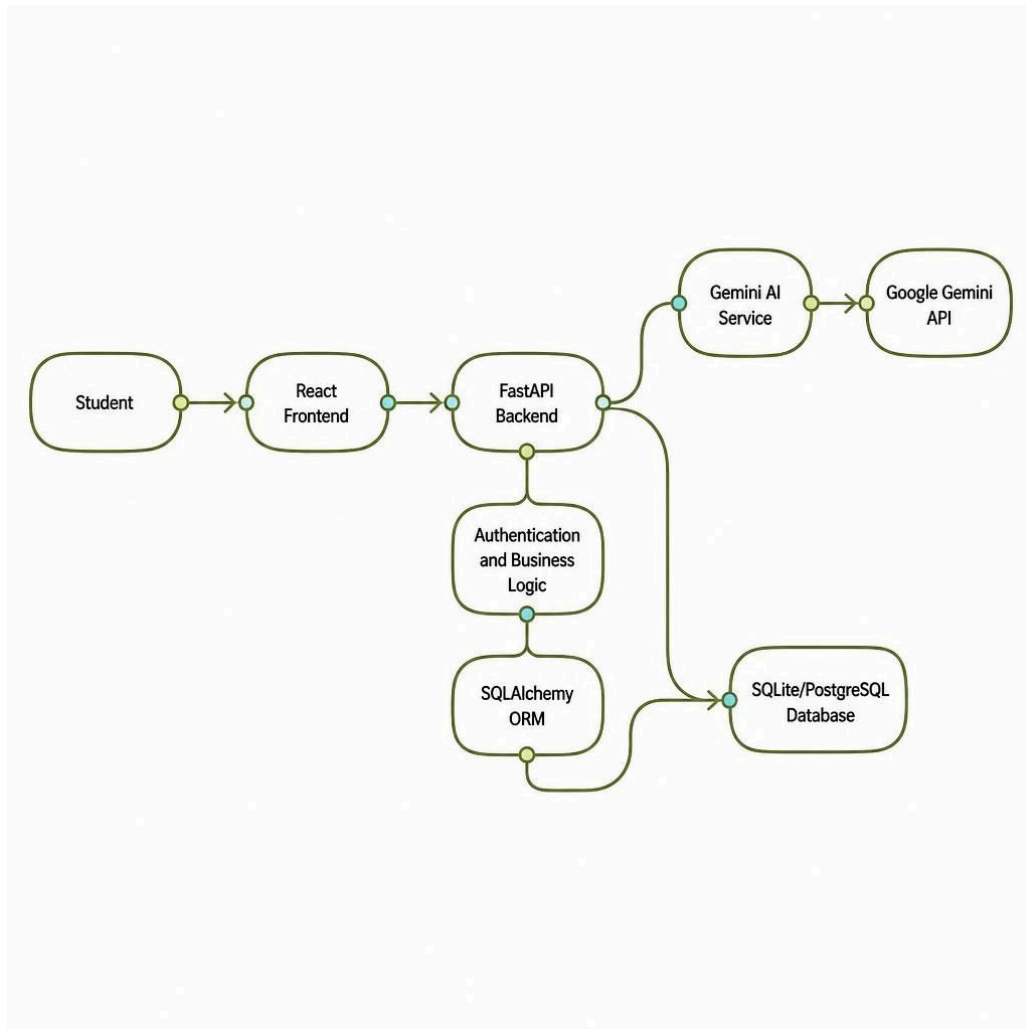
Generate topic-based quizzes automatically



Progress & Analysis

Track performance, identify weak topics, get recommendations

How It All Fits Together



Architecture Highlights

The system follows a clean **client-server model** with a clear separation of concerns.

→ **React Frontend** — Responsive UI built with TypeScript and Tailwind CSS

→ **FastAPI Backend** — RESTful API with JWT authentication and business logic

→ **Gemini AI Service** — Abstraction layer for all AI-powered features

→ **Database Layer** — SQLite (dev) / PostgreSQL-compatible for production

TECHNOLOGY STACK

Built With Modern, Proven Tools



Backend

Python · FastAPI · SQLAlchemy · Pydantic



Database

SQLite (development) · PostgreSQL-compatible architecture



Authentication

JWT tokens · bcrypt password hashing



Artificial Intelligence

Google Gemini API for all AI-powered features



Frontend

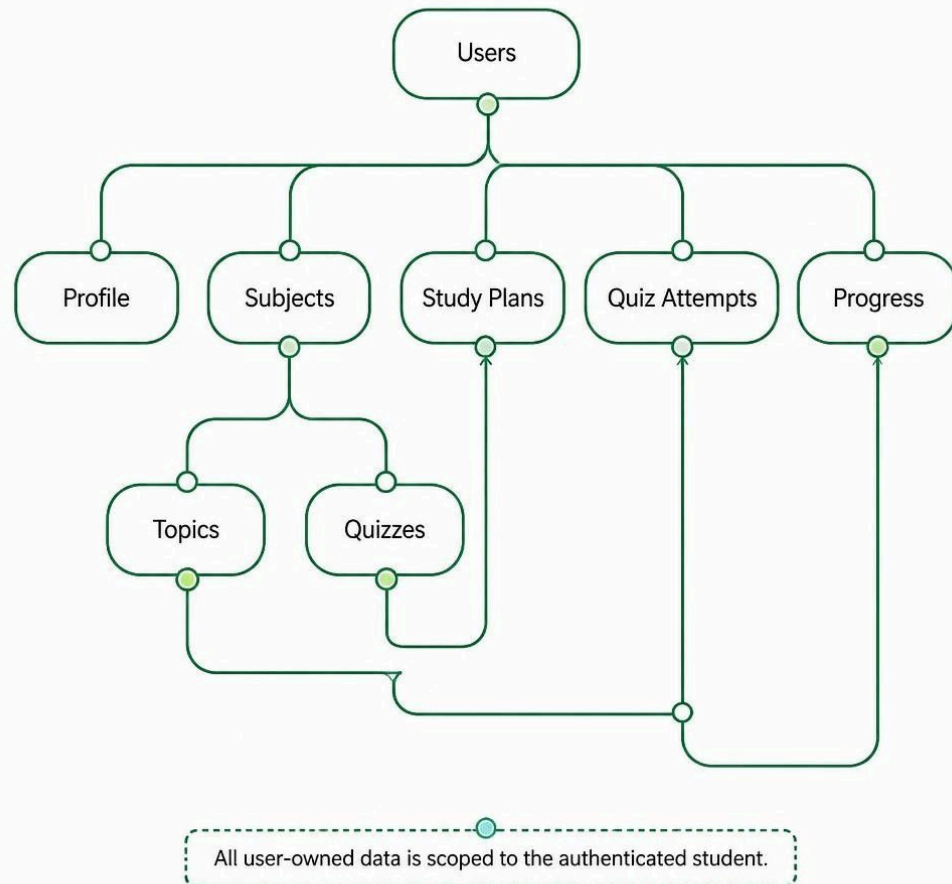
React · TypeScript · Vite · Tailwind CSS



Testing

Pytest · API testing for backend validation

Data Model & Entity Relationships



Core Entities

Users & Profile

Authentication credentials, learning preferences, and education level

Subjects & Topics

Organized content hierarchy for each student's curriculum

Study Plans & Quizzes

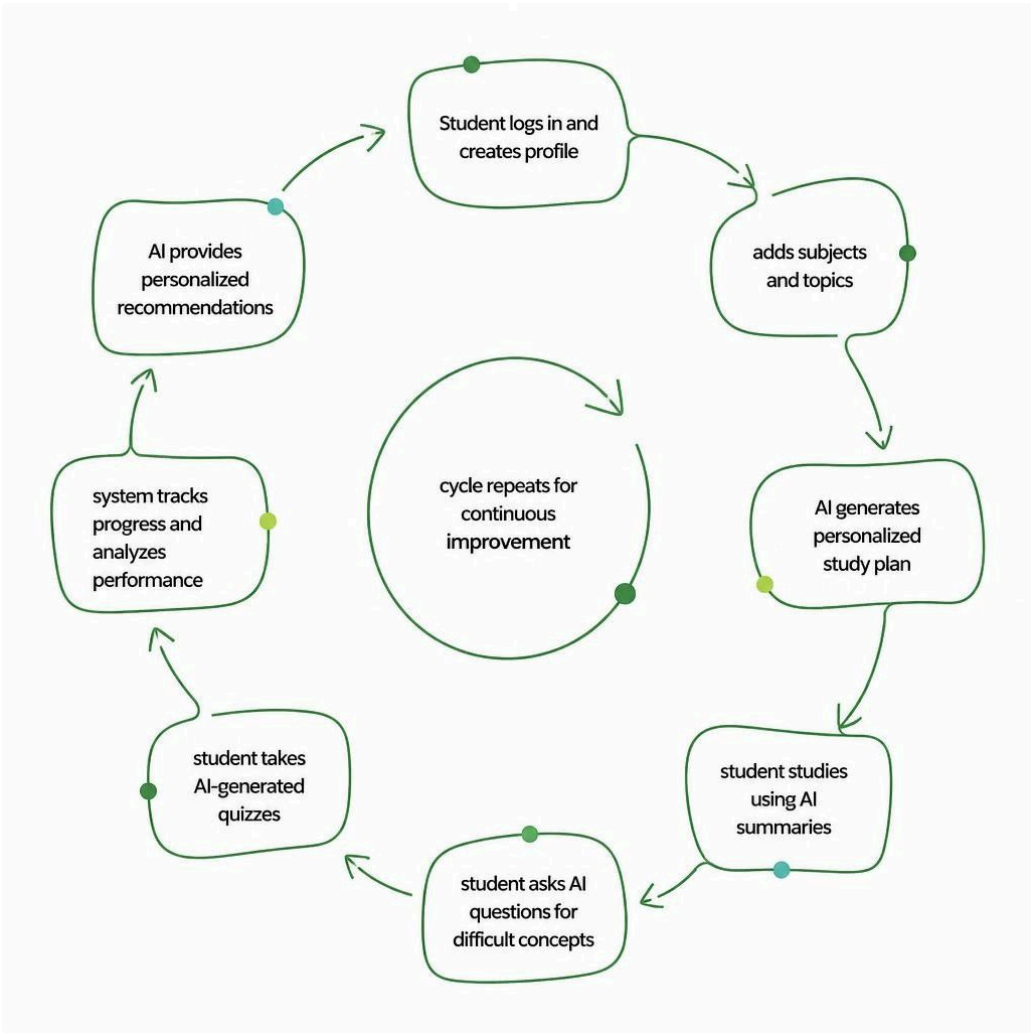
AI-generated plans and auto-generated quizzes linked to topics

Quiz Attempts & Progress

Scores, timestamps, and performance trends per topic

Ownership relationships ensure students can only access their own data — a critical security and privacy design principle.

How Students Use AI StudyBuddy



End-to-End Learning Flow

From onboarding to mastery, every step is guided by AI — creating a continuous loop of learning, practice, and improvement.

01

Onboard & Organize

Set up profile, add subjects and topics

02

Plan & Study

AI generates schedule; student uses summaries and Q&A

03

Practice & Assess

Take AI-generated quizzes with instant scoring

04

Analyze & Improve

AI identifies weak areas and recommends next steps

 **What Makes It Innovative:** Unified AI-powered workflow, personalized adaptive planning, automated quiz generation and scoring, intelligent performance analysis, and secure multi-user architecture — all in one platform.

Summary & Future Vision

What We Built

A full-stack AI-powered student learning platform with authentication, subject management, AI study planning, summarization, Q&A, quiz generation, progress tracking, and personalized recommendations.

Impact

Helps students study smarter, save time, improve understanding, and track progress — all in one place.

Future Enhancements

Mobile app, collaborative study rooms, voice-based AI interaction, LMS integration, multilingual support, and advanced analytics.

AI StudyBuddy is a complete, scalable, and intelligent student learning platform designed to modernize how students study and succeed.

Team 11: Alan (Team Leader) · Mathew · Karthikeyan · Kishore

Thank You!